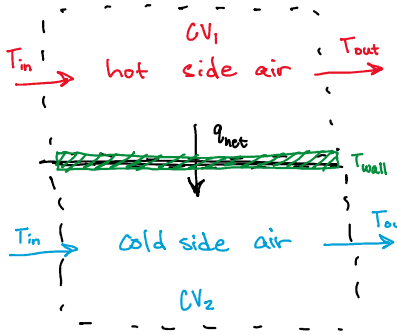


energy balance

for CV1: $\dot{m}\left(h+\frac{V^2}{2}+gz\right)_{in} - \dot{m}\left(h+\frac{V^2}{2}+gz\right)_{out} + \dot{q} - \cancel{\dot{W}} = 0$

steady state: $\cancel{\dot{q} = \dot{m}c_p(T_{out}-T_{in})}$ $\dot{q}_{hot} = \dot{q}_{cold} = \dot{q}_{wall}$

no shaft/pump: $\dot{W}=0, \Delta p = 0$

unsure
- ΔKE/PE ≈ 0 ?
- assume ideal gas of air → air (c_p const?) & $\Delta h = c_p \Delta T$

→ simplify: $\dot{m}(h_{in}-h_{out}) = \dot{q}$!

$\boxed{\dot{q}_{hot} = \dot{m}c_p(T_{out}-T_{in})}$

avg rate flux $\dot{q} = q'' A$ [W] $\left[\frac{W}{m^2}\right]$ is it a cylinder or plate?

"same" $\dot{m}(h_{in}-h_{out}) = \dot{q}$

$\boxed{\dot{q}_{cold} = \dot{m}c_p(T_{out}-T_{in})}$

for T_{wall}: convection $\dot{q} = hA(T_s - T_b)$

$\dot{q} = hA(T_b - T_{wall_{hot}}) \rightarrow T_b \approx \frac{T_{in}+T_{out}}{2}$ (bulk free stream rule)

$\dot{q} = hA(T_{wall_{cold}} - T_b)$

link if we assume non uniform thru plate

conduction

T_{hot}

T_{cold}

can we assume $T_{wall\,top} = T_{wall\,bottom} = T_{wall}$?

before: flat wall only $\boxed{\dot{q} = \frac{kA}{t}(T_{hot}-T_{cold})}$

$T_{wall_{hot}}$ $T_{wall_{cold}}$

★ if we assume T_{wall} constant for thin plate this one might cancel out ★

combine: for total transfer $\dot{q}_{net}$ AND ($\dot{q}_{hot} = \dot{q}_{cold}$)

$\dot{m}c_p(T_{out}-T_{in}) = hA(T_b - T_{wall})$

also equal!

$h = \frac{\dot{m}c_p(T_{out}-T_{in})}{A(T_b - T_{wall})}$ NO! h from Nusselt

$\dot{m}c_p(T_{out}-T_{in}) = hA(T_b - T_{wall})$

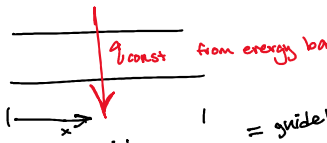
$h = \frac{\dot{m}c_p(T_{out}-T_{in})}{A(T_{wall}-T_b)}$

Re → Nu → h same as before?

knowns: $\dot{m}$, c_p, $A = lwh$?, h (by Re), T_{in}

unknowns: T_{out}, ~~T_{in}~~, ~~h~~, T_{wall} → can get cancelled if uniform!

★ all $\dot{q}$ are equal ★



from energy balance

= guidelines

$\int \frac{\Delta q}{\Delta L} dL \quad q = \dot{m}c_p(T_{out}-T_{in})$ ← variable

$\int_0^L \frac{\dot{m}c_p \Delta T}{L} dL$

↳ $\dot{m}$

$\dot{m}c_p\Delta T\left[\ln(L)\right]_0^L$

$\dot{m}c_p\Delta T\left(\ln(L) - 0\right)$

$$q = \frac{kA}{t}\left[\left(\frac{q_h}{h_h A} + \frac{T_{h1}+T_{h2}}{2}\right) - \left(\frac{q_c}{h_c A} + \frac{T_{c1}+T_{c2}}{2}\right)\right]$$

$$\frac{kA}{t}\frac{q_h}{h_h A} + \frac{kA}{t}\left(\frac{T_{h1}+T_{h2}}{2}\right) - \left(\frac{kA}{t}\frac{q_c}{h_c A}\right) - \left(\frac{kA}{t}\frac{T_{c1}+T_{c2}}{2}\right)$$

$$q + \frac{kA}{t}(q_h + q_c) = \frac{kA}{t}\left(\frac{T_{h1}+T_{h2}}{2} - \frac{T_{c1}+T_{c2}}{2}\right)$$

$h_c A$

$$\boxed{q = \frac{kA}{t}\left(\frac{T_{h1}+T_{h2}}{2} - \frac{T_{c1}+T_{c2}}{2}\right)}$$

$$T_{h_2}: \quad \frac{T_{h1}+T_{h2}}{2R} - \frac{T_{c1}+T_{c2}}{2R} = \dot{m}c_p(T_{h1}-T_{h2})$$